CARLOS ROMERO MADRID

Computational Physicist · Data Science · HPC Software Engineer
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✦ LIVE INTERACTIVE WEB DOSSIER & DIGITAL EXPERIENCES
I have created an online and interactive CV for the DAV group, hope you like it! → https://carrommad.github.io/cool-cv

EDUCATION

BSc Physics & BSc Mathematics
University of Seville, 2013–2018
8.35/10 · 3rd of Class

MSc Plasma & Fusion Physics
Univ. Aix-Marseille, 2019–2020
15.43/20 · 1st of Class

MSc Advanced Mathematics
UNED 2023-Present (ABD)

TECHNICAL SKILLS

Languages
- **Advanced**
  Fortran, C++, Python, Shell, Git, OpenMP, MPI
- **Intermediate**
  OpenACC, CUDA, oneAPI

HPC & Systems
- **Large-scale cluster deployment**: MN5, Leonardo, LUMI
- **Parallelisation**: OpenMP, MPI, PETSc

PROFILE

Physicist and software engineer with 5+ years of experience spanning plasma and charged-particle simulation, HPC system engineering, and high-performance scientific computing in international research institutions. Includes 1-2 years as a researcher at the University of Seville developing particle-in-cell (PIC) and fluid plasma models, building a solid physical grounding in charged-particle and plasma behavior alongside the numerical methods used to simulate it.

Experienced in deploying and operating large-scale distributed workloads on supercomputing clusters (MareNostrum5, Leonardo, LUMI), parallelizing production simulation codes with MPI/OpenMP/CUDA, and building data pipelines and ML systems in Python, the kind of theory-to-implementation work needed to take a physical model from equations to a running, production-grade solver.

Solid software engineering practices: code reviews, technical documentation, and cross-disciplinary teamwork with physicists and engineers. Actively integrates AI-assisted development tools to accelerate productivity and code quality.

PROFESSIONAL EXPERIENCE

Barcelona Supercomputing Center (BSC)
Barcelona
HPC Scientific Software Engineer · Performance Team
May 2025 – Present

- Contributing to ocean–atmosphere coupling in the DestinationE and ClimateDT digital twin projects, working on scalable model integration across HPC infrastructure.
- Developing new capabilities in the ECLand climate model, collaborating with the Climate Variability and Change team on production-grade scientific services.
- Deploying, configuring and monitoring long-running services and simulation workflows on MareNostrum5, ensuring reliability and performance at scale.

– **GPU offloading**: CUDA, OpenACC, oneAPI
– **Performance**: Job scheduling & profiling

Data & ML

– Python data analysis
– Predictive maintenance & classification
– Exploratory analysis & diagnostics

AI-assisted Dev

– DeepSeek: code generation & review
– Warp & Antigravity terminals

Collaboration

– Git / GitHub workflows
– Code reviews & technical docs
– Cross-disciplinary team experience

LANGUAGES

– **Advanced**
Spanish, English, French
– **Intermediate**
Italian, German

Research Engineer · Fusion Team *May 2023 – April 2025*

- Member of the Advanced Computing Hub project: optimised parallelisation of nuclear fusion simulation codes using OpenMP, MPI, PETSc, oneAPI and OpenACC on European Tier-0 supercomputers.
- Achieved measurable performance improvements in distributed multi-node workloads; participated in code reviews and maintained technical documentation for the team.
- Enhanced simulation features in the MAGNET module, implementing electromagnetic modelling for fusion superconducting magnets as part of the ALYA multi-physics initiative.
- Collaborated closely with applied scientists and engineers across disciplines, presenting trade-offs and results to both technical and non-technical stakeholders.

National Renewable Energy Centre
Pamplona

Data Scientist · Analysis and Design *May 2022 – September 2022*

- Built Python-based data pipelines for real-time characterisation, optimisation and control of offshore wind plants.
- Optimised predictive maintenance algorithms using pandas and scikit-learn; designed expert systems for automated turbine data classification.
- Delivered analysis-driven insights to engineering teams, bridging data science outputs with operational decisions.

University of Seville
Seville

Research Scientist · Plasma & Fusion Technology *September 2020 – April 2022*

- Conducted theoretical and computational plasma physics research on HPC systems using gyrokinetic theory and MHD simulations.
- Developed Python diagnostics for 3D particle tracking and Fortran simulations of magnetic equilibria.
- Lectured undergraduate Quantum Physics courses; presented research findings at group meetings and scientific events.

Fraunhofer Institute FHR
Bonn

Research Scientist · Cognitive Radar *August 2018 – April 2019*

- Implemented simulated annealing optimization in MATLAB for non-planar radar antenna design, improving directivity and accuracy for automotive applications.